PRODUCTION ENGINEERING REPORT

Parthsaarthi

Production Resilience & Booking-Integrity Validation

SIP Mentor Booking Platform · Indian Institute of Management Lucknow
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CONFIDENTIAL — LEADERSHIP REVIEW

1. Purpose

Two classes of test were conducted against the live production environment to validate the platform's readiness for full-cohort, high-concurrency usage — specifically the conditions expected during a placement-season booking window, when large numbers of students act on the system simultaneously.

- **Infrastructure load testing** — does the system remain available and responsive as concurrent users scale from 1 to 800?
- **Data-integrity stress testing** — when many students compete for the same limited mentor slots at the exact same moment, does the system guarantee that each slot is awarded to exactly one student, with no double-booking?

2. Test 1 — Infrastructure Load Capacity

Concurrent request volume was ramped through 1, 5, 10, 25, 50, 100, 200, 400, 600, and 800 simultaneous users, against both the public application shell and an authenticated, database-backed API call representative of real usage (mentor browsing).

> **Result: PASS** — zero failures recorded at every concurrency tier.

Concurrent Users	Application Shell	Authenticated API (DB-backed)
100	100 / 100 succeeded · 23ms median	**100 / 100 succeeded · 104–183ms**
400	400 / 400 succeeded · 85ms median	**400 / 400 succeeded · 346–605ms**
800	800 / 800 succeeded · 85ms median	**800 / 800 succeeded · 738ms–1.26s**

No errors, no service interruption, and no abnormal resource consumption were observed at any tier. The application server returned to baseline CPU and memory immediately after each test. The system comfortably absorbed a simultaneous load roughly **2–3x the entire student cohort size** with no degradation in availability.

3. Test 2 — Booking Integrity Under Concurrent Demand

This test simulated the highest-risk real-world scenario for a booking platform: a large number of students attempting to book a small number of mentor slots in the same instant — the exact condition expected when a popular mentor's slots open. Two scenarios were run:

- **Single hot slot** — 300 students competing simultaneously for 1 available session.
- **Distributed load** — a mentor releasing 10 slots, with 300 students competing simultaneously and randomly across all 10.

3.1 Initial Finding — Confirmed Data-Integrity Defect

> In the single-slot scenario, **7 students were simultaneously confirmed into a session designed for 1**, due to a timing gap in the original booking logic under heavy concurrent load. This was independently confirmed directly against the production database — not merely application responses — establishing it as a genuine, reproducible defect rather than a testing artifact.

3.2 Remediation

The booking-claim logic was redesigned to use a single, database-enforced atomic operation, ensuring the database itself — not just the application layer — guarantees a slot can never be awarded more times than its configured capacity allows, regardless of how many requests arrive at the same instant.

3.3 Post-Remediation Validation

> **Result: PASS** — verified in both scenarios, directly against production database records.

Scenario	Concurrent Demand	Capacity	Successful Bookings	Outcome
Single hot slot	300 students	1	Exactly 1	**299 correctly declined**
Distributed (10 slots)	300 students	10	Exactly 10	**1 per slot · 290 correctly declined**

No slot, in any test, was awarded more times than its configured capacity after remediation. All test data (throwaway mentor and slot records used to conduct these tests) was fully removed from production following validation, with zero residual footprint and no impact to real student or mentor data.

4. Secondary Finding

During load testing, an unrelated, pre-existing defect was also surfaced and corrected: the student-facing "browse all mentors" view was failing for any session that included a mentor without committee/AIG affiliation (e.g. peer mentors) — a configuration introduced in a prior release. This has been fixed and deployed; the feature is now fully functional.

5. Overall Assessment

Dimension	Status
Availability under load (up to 800 concurrent)	**Confirmed resilient**
Booking data integrity under concurrent demand	**Defect found, fixed, and validated**
Production impact of testing	**None — all test artifacts fully removed**

The platform is validated as ready to handle full-cohort concurrent usage, including the highest-contention scenario of a popular mentor's slots opening to the entire student body at once. The concurrency defect identified during this exercise represented a real risk to session-booking integrity and has been remediated and independently re-verified before this report was issued.